

Sacituzumab govitecan in Breast Cancer

1. Drug Summary

Sacituzumab govitecan (brand name Trodelvy) is a first-in-class antibody-drug conjugate (ADC) composed of a humanized IgG1 anti-TROP-2 monoclonal antibody linked to SN-38 (the active metabolite of irinotecan) ¹ ². It is approved for metastatic triple-negative breast cancer (TNBC) after ≥ 2 prior therapies, and recently also for heavily pretreated hormone receptor-positive/HER2-negative (HR+/HER2-) breast cancer ³ ⁴. In pivotal trials, SG markedly improved outcomes: in TNBC (ASCENT trial) it raised median progression-free survival (PFS) from ~1.7 to 5.6 months and overall survival (OS) from ~6.7 to 12.1 months vs chemotherapy ⁵ ⁶. In pretreated HR+/HER2- patients (TROPICS-02 trial), SG extended median PFS and OS (e.g. OS 14.4 vs 11.2 months) ⁷ ⁸. The major toxicities reflect its payload and antigen: myelosuppression (neutropenia ~61%, grade ≥ 3 in ~47%) and diarrhea (~65%, grade ≥ 3 in ~12%) are common ⁹ ¹⁰, and it carries boxed warnings for severe neutropenia and diarrhea ¹¹. Hypersensitivity to sacituzumab (Trodelvy) is a contraindication ¹².

2. Identifiers & Synonyms

- **DrugBank ID:** DB12893 ¹³.
- **ChEMBL ID:** ChEMBL3545262.
- **Brand Name:** Trodelvy (sacituzumab govitecan-hziy) ¹⁴.
- **Synonyms/Codes:** *sacituzumab govitecan*, sacituzumab govitecan-hziy, hRS7-SN38, RS7-SN38, IMMU-132 (IMMU 132), RS7SN38 ¹⁴. These refer to the same ADC. (No small-molecule analog exists; SN-38 itself is the payload.)

3. Mechanism of Action (Breast Cancer Context)

Sacituzumab govitecan targets **Trop-2** (tumor-associated calcium signal transducer 2, gene *TACSTD2*), an epithelial cell surface antigen frequently overexpressed in breast cancers (especially TNBC) ¹. The antibody binds TROP-2 on tumor cells, triggering rapid internalization of the ADC ¹⁵. Once internalized, the hydrolysable linker releases SN-38 intracellularly ¹⁵. SN-38 inhibits DNA topoisomerase I, stabilizing the topoisomerase-DNA cleavage complex and causing DNA single- and double-strand breaks, ultimately leading to cell cycle arrest and apoptosis ¹⁵. (Secondary pathways such as p53-mediated checkpoint activation and caspase-driven apoptosis are engaged by the ensuing DNA damage.) Notably, SN-38 may also disrupt binding of *Far Upstream Element-Binding Protein 1* (FUBP1) to oncogene promoters, potentially affecting transcription ¹⁶. In addition, some ADCC activity may occur: the Fc portion of the humanized IgG1 can engage Fc γ receptors on immune cells, contributing to antibody-dependent cellular cytotoxicity ¹⁷. Separately, Trop-2 itself signals to promote tumor growth: it activates PKC/MAPK and β -catenin pathways (upregulating cyclin D1, c-Myc) and integrin/Src/FAK signaling (promoting migration) ¹⁸. Thus, SG blocks Trop-2-driven proliferation/motility while delivering a potent DNA-damaging payload.

4. Primary Targets (HGNC symbols, human only)

- **TACSTD2 (HGNC:11278)** – *Tumor-associated calcium signal transducer 2*, i.e. Trop-2 antigen ¹⁹ . (Targeted by the antibody component of SG.)
- **TOP1 (HGNC:12022)** – *DNA topoisomerase I*, the intracellular target of SN-38 ¹⁵ . (Inhibited by SN-38, causing DNA damage.)
- **FUBP1 (HGNC:3926)** – *Far upstream element-binding protein 1*. SN-38 may also inhibit FUBP1–DNA interactions as a secondary effect ¹⁶ . (Though not a classic “target”, it is impacted by the drug.)

No other human proteins are specifically known targets of sacituzumab govitecan. (Metabolic enzymes such as UGT1A1 influence SN-38 clearance but are not direct drug “targets.”)

5. Pathways (Overview)

Sacituzumab govitecan engages multiple cellular pathways, chiefly those involving Trop-2 signaling and DNA damage response. **Trop-2-associated pathways** include adhesion and proliferation networks: e.g. focal adhesion (cell–matrix interactions), MAPK signaling, and β -catenin/Wnt signaling ¹⁸ ²⁰ . **DNA damage and cell cycle pathways** are key: SG induces double-strand DNA breaks, engaging p53, ATM/ATR, and cell-cycle checkpoint cascades (G1/S and G2/M arrest), as well as apoptosis pathways. Relevant Reactome/KEGG pathways include DNA replication (hsa03030), cell cycle (hsa04110), p53 signaling (hsa04115), and general “Pathways in cancer” (hsa05200/05224).

Immune pathways may also be involved: as an IgG1 ADC, Fc γ receptor signaling and ADCC-related pathways can contribute to cell kill. (Hallmark gene sets suggest luminal tumors often upregulate focal adhesion pathways, whereas basal/TNBC tumors upregulate inflammation and WNT pathways ²⁰ .) In sum, SG’s effects span cell adhesion/signaling networks (via Trop-2) and core cytotoxic pathways (DNA damage response, apoptosis).

6. Upregulated pathways (by subtype/context)

- **HR+/HER2– (Luminal A/B):** These tumors often upregulate hormone-response and proliferation signatures. For example, Luminal A subtypes characteristically show enrichment of focal adhesion pathways (cell–matrix interactions) ²⁰ . Luminal B tumors tend to have higher cell-cycle/E2F target activity.
- **HER2+:** These cancers frequently have upregulated ERBB2/HER2 signaling, PI3K/AKT, and MAPK pathway activity, driving rapid growth. Immune signatures are variable.
- **Triple-negative (Basal-like/TNBC):** High expression of inflammatory and proliferative pathways is typical. MSigDB data note that basal/TNBC tumors have upregulated WNT signaling ²⁰ . Other upregulated features in TNBC include TNF- α /NF κ B, IL6/JAK/STAT3, and IFN γ -response pathways, along with strong cell-cycle (E2F, G2M) signatures.
- **gBRCA1/2-mutated and HRD+ tumors:** These often overlap with TNBC and show high replication stress. Homologous recombination pathways (if intact) may be active, but by definition HRD tumors lack effective HR repair. They frequently upregulate alternative DNA damage responses and cell-cycle checkpoint pathways (e.g. ATR/CHK1) to cope with genomic instability.

7. Downregulated pathways (by subtype/context)

- **HR+/HER2-:** Compared to TNBC, luminal tumors show lower immune/inflammatory pathway activity (e.g. lower IFN γ and IL6 signaling) and generally lower WNT/EMT signatures.
- **HER2+:** These tumors may have downregulated estrogen-receptor pathways (often mutually exclusive with strong HER2 signaling) and variable immune signatures (some HER2+ subgroups have low lymphocyte infiltration).
- **Triple-negative:** TNBC by definition lacks estrogen and HER2 signaling, so those pathways are down. Some TNBC subtypes downregulate normal epithelial differentiation programs.
- **gBRCA/HRD+:** By definition, HR repair pathways are downregulated/deficient. Non-homologous end joining may be up, but classical BRCA-mediated repair is suppressed.

(In general, “downregulated” subtypes have the converse of the upregulated pathways above.)

8. Sensitivity pathways for upregulation (prioritized)

Upregulation of certain pathways or factors can increase SG efficacy (sensitivity):

- **TROP-2 (TACSTD2) expression:** High cell-surface Trop-2 is critical. Tumors with strong TACSTD2 upregulation bind and internalize more ADC, increasing payload delivery. Indeed, one case report found that a patient whose tumor had *no* TROP-2 expression exhibited rapid progression, whereas another patient with *focal amplification* of TACSTD2 had a deep, durable response ²¹. Thus, upregulated Trop-2 is strongly sensitizing.
- **TOP1 activity:** High levels of active topoisomerase I mean more SN-38 target. Intact TOP1 enables efficient SN-38 binding and DNA cleavage. (Conversely, acquired TOP1 mutation E418K abolished SN-38 activity ²².) Therefore, upregulated TOP1 expression or activity sensitizes cells.
- **Cell proliferation / S-phase activity:** Although not directly cited, rapidly cycling cells (with elevated E2F and DNA replication pathways) are known to be more susceptible to topoisomerase inhibitors. Upregulation of cell-cycle genes thus generally enhances sensitivity to DNA-damaging payloads.
- **Apoptotic and DNA damage signaling:** Upregulation of intrinsic apoptosis pathways (e.g. active caspases, functional p53) should promote killing once DNA damage occurs. (No direct citation, but biologically, cells primed for apoptosis die more easily from SN-38–induced damage.)
- **Immune effector pathways (ADCC):** Overexpression of Fc γ receptors or activation of NK cells/macrophages can increase ADC-mediated killing. High immune activity can thus sensitize tumors to the antibody component. Indeed, some ADCC activity is associated with sacituzumab treatment ¹⁷.

9. Sensitivity pathways for downregulation (prioritized)

Conversely, downregulation of certain pathways can heighten SG sensitivity:

- **Drug metabolism down:** If *UGT1A1* activity is low, SN-38 is not efficiently glucuronidated and inactivated. Downregulation of UGT1A1 (or presence of the low-activity UGT1A128 variant) *leads to higher SN-38 exposure and often greater tumor kill (at the risk of toxicity).* (The label notes UGT1A1 28/28 status increases neutropenia risk ²³.)
- **Efflux transporter down:** Reduced expression of efflux pumps (e.g. ABCG2/MDR1) would keep SN-38 inside cells, increasing efficacy.
- **DNA repair down:** Cells deficient in DNA repair (e.g. HR deficiency) cannot fix SN-38–induced breaks, so they are more easily killed. Thus downregulation of homologous recombination or checkpoint repair pathways heightens sensitivity.

- *Survival signaling down:** Reduced activity of pro-survival pathways (e.g. PI3K/AKT) or anti-apoptotic proteins (e.g. BCL2) would make cells more prone to SN-38-triggered death.

(These are conceptual: human evidence is strongest for UGT1A1 and Trop-2 levels.)

10. Resistance pathways for upregulation (prioritized)

Tumors can acquire resistance by upregulating pathways that mitigate SG's action:

- **Drug efflux/metabolism up:** Overexpression of drug pumps (e.g. ABCG2/BCRP, ABCB1/MDR1) or detoxifying enzymes (UGT1A1) leads to rapid SN-38 removal. In particular, upregulated UGT1A1 activity glucuronidates SN-38 to its inactive form ²⁴, directly conferring resistance.
- **DNA repair up:** Enhanced repair of DNA breaks (upregulated ATM/ATR, BRCA2, etc.) can blunt SN-38's damage. For example, any restoration of homologous recombination would let cells survive what SG induces.
- **TOP1 mutation:** Upregulation (selection) of mutant TOP1 enzymes can prevent SN-38 binding. Indeed, one tumor developed a TOP1 E418K mutation in resistant clones, which fully abrogated SN-38-induced DNA breaks ²².
- **TROP-2 loss or mutation:** Upregulation of mechanisms that reduce TROP-2 on the surface (for instance, mutations affecting its localization) abrogate ADC binding. In a resistant case, a mutation TROP-2^ΔT256R emerged that prevented membrane localization; patients with such altered/low Trop-2 had poor response ²².
- **Survival/proliferation pathways up:** Activation of alternative growth signals (e.g. PI3K/AKT, NFκB) or cell-cycle checkpoints can allow tumor cells to withstand DNA damage.

11. Resistance pathways for downregulation (prioritized)

Resistance can also arise if certain pathways are suppressed:

- **Trop-2 downregulation:** If TACSTD2 expression is lost or severely reduced ("downregulated"), SG cannot bind its target and the drug is ineffective ²¹.
- **Apoptosis down:** Loss of apoptosis mediators (e.g. p53 mutation, downregulation of caspases) would allow cells to tolerate SN-38 lesions.
- **Cell-cycle quiescence:** If tumor cells slow cycling (downregulate E2F targets), they may avoid SN-38's S-phase toxicity.
- **Immune evasion:** Downregulation of Fcγ receptors or HLA molecules could blunt ADC and immune effects, contributing to resistance.

(Again, concrete human evidence is limited here; most resistance pathways are inferred from observed mechanisms and related chemotherapy experience.)

12. Breast cancer subtype-stratified evidence

TNBC: Sacituzumab govitecan is *standard of care* in metastatic TNBC after prior therapy. The phase III ASCENT trial (NCT02574455) demonstrated significantly better outcomes vs chemotherapy: median PFS 5.6 vs 1.7 months (HR 0.41) and OS 12.1 vs 6.7 months (HR 0.48) ⁵ ⁶. The FDA (April 2020) and EMA (Nov 2021) approved SG for this setting ³ ²⁵. The US NCCN guidelines now recommend SG as a preferred Category 1 option for metastatic TNBC after ≥2 prior regimens. The UK NICE TA819 similarly approves it for

unresectable TNBC after ≥ 2 therapies ²⁶. Common biomarkers (Trop-2 IHC) have not been required for use, since most TNBCs express Trop-2.

HR+/HER2-: SG was recently shown effective in endocrine-resistant luminal disease. The phase III TROPiCS-02 trial (NCT03901339) in heavily pretreated HR+/HER2- MBC reported improved PFS and OS: e.g. OS 14.4 vs 11.2 months (HR 0.79) ⁸. These data led to FDA approval in Feb 2023 for HR+/HER2- MBC after endocrine and ≥ 2 chemotherapies ⁴, and EMA approval as noted above ²⁵. Regulatory guidance now includes SG as a late-line option in HR+ disease; NCCN updated its recommendations accordingly. As of now, SG is **not** indicated for HER2+ breast cancer, and clinical trials of SG (with/without trastuzumab) in HER2+ tumors are ongoing.

gBRCA1/2-mutated and HRD: There is no specific label for SG in BRCA-mutant or HRD breast cancer, which are often TNBC or luminal. However, preclinical rationale suggests SG could be effective in HRD contexts (similar to platinum or PARP inhibitors). For example, one study on anti-Trop-2-SN-38 ADCs noted that BRCA1-mutant TNBC (basal-like) may be sensitive due to synthetic lethality with DNA damage ²². Clinical data are sparse, though trials are exploring combinations of SG with DNA repair-targeting agents (PARP inhibitors) in HRD tumors. Currently, PARP inhibitors are preferred for BRCA-mutant MBC, but SG is a consideration especially if PARPi fails.

13. Contraindications and Safety (label/guideline)

Sacituzumab govitecan (Trodelyv) is contraindicated in patients with a history of severe hypersensitivity to the drug ¹². Major safety concerns (from FDA/EMA labels) include: **neutropenia** (grade ≥ 3 neutropenia in ~47% of patients ⁹), which can be life-threatening; **diarrhea** (any grade in ~65%, grade ≥ 3 in ~12% ¹⁰); and infusion-related reactions/anaphylaxis. These are captured in boxed warnings for neutropenia and diarrhea ¹¹ ²⁷. Management guidelines advise routine CBC monitoring and dose delays or G-CSF support for neutropenia, and prompt antidiarrheal therapy (e.g. loperamide) for diarrhea. UGT1A1*28 homozygosity (poor metabolizer) is a pharmacogenomic risk factor for neutropenia ²³, so dose adjustment or monitoring is recommended for these patients. Other precautions: SG can cause fetal harm, so avoid in pregnancy and use effective contraception ¹¹. Embryo-fetal toxicity is noted on the label. Contraindications beyond hypersensitivity are not listed. Overall, SG's safety profile reflects typical topoisomerase-I-mediated toxicity and ADC infusion risks, and guidelines emphasize established supportive measures for these effects.

14. Trial and guideline context

Key clinical trials include the first-in-human Phase I/II studies (IMMU-132-01) and the pivotal Phase III **ASCENT** study in TNBC ⁶ ⁵, and the Phase III **TROPiCS-02** trial in HR+/HER2- MBC ⁷ ⁸. These trials underlie current approvals. Regulatory actions: FDA granted accelerated approval for TNBC in 2020 (full approval in 2021) and expanded to HR+ in 2023 ⁴; EMA granted approval for both indications in 2021 ²⁵. In practice, US and international guidelines now list SG for advanced TNBC (category 1 NCCN, NICE-approved) and for late-line HR+ breast cancer. For example, the UK NICE TA819 (Aug 2022) explicitly recommends SG for advanced TNBC after ≥ 2 therapies ²⁶. The US NCCN 2024 guidelines similarly include SG for TNBC and HR+ settings (preferred category 1 for TNBC, and category 2A for HR+ after ≥ 3 regimens). Ongoing research includes trials of SG in earlier lines (e.g. first-line TNBC), combinations with immunotherapy (e.g. SG + pembrolizumab showed improved PFS in early reports), and investigational use in other Trop-2-expressing cancers (e.g. urothelial, lung).

15. Additional Mechanistic/Clinical Notes

- **Bystander effect:** SN-38 has moderate membrane permeability; thus, after Trop-2-mediated release, SN-38 may diffuse to neighboring tumor cells (even Trop-2-negative ones) and kill them. This bystander effect could contribute to efficacy in heterogeneous tumors.
- **Antibody effector function:** In vitro evidence suggests the RS7 antibody component retains some ADCC activity ¹⁷, which may help clear Trop-2⁺ cells.
- **Immunogenicity:** As a protein therapeutic, SG can induce anti-drug antibodies in rare cases. There is no standard immunogenicity monitoring, but clinicians should watch for infusion reactions.
- **Pediatric use:** SG is not approved for children; its safety in younger populations has not been established.
- **Pharmacokinetics:** The ADC has a half-life ~16 hours and delivers a relatively high SN-38 exposure. Most SN-38 in serum remains antibody-bound ²⁸. Uridine diphosphate glucuronosyltransferase activity (UGT1A1) is the main SN-38 clearance pathway ²⁹.
- **Comparative context:** Sacituzumab govitecan is the first Trop-2-directed ADC approved. It is distinct from other breast ADCs (e.g. HER2-targeted trastuzumab deruxtecan) by its different target and payload. Unlike highly cytotoxic payloads (e.g. microtubule inhibitors), SN-38's topoisomerase mechanism may have unique synergy with DNA repair defects.

Pathway Evidence Table: Key pathways modulating SG sensitivity or resistance are listed below.

Pathway (ID/Name)	Regulation (Up/Down)	Effect on SG (Sensitive/Resistant)	Rationale	References
Trop-2 (TACSTD2) signaling	Upregulated	Sensitive	High Trop-2 expression increases ADC binding/internalization ²¹ .	²¹
Topoisomerase I (TOP1) activity	Upregulated	Sensitive	SN-38 targets TOP1-DNA complex; active TOP1 is required for DNA cleavage ¹⁵ .	¹⁵
UGT1A1-mediated metabolism	Upregulated	Resistant	UGT1A1 glucuronidates (inactivates) SN-38 ²⁴ , lowering active drug.	²⁴
FcγR (ADCC/ Immune effector)	Upregulated	Sensitive	ADC Fc engages Fcγ receptors, promoting immune cell cytotoxicity ¹⁷ .	¹⁷

Sources: DrugBank ^{13 14 2 15}; FDA/EMA labels ^{12 25}; pivotal trials ^{5 8 7}; Mechanistic studies ^{30 18}; guideline summaries (NICE) ²⁶. All data are human-specific and derived from authoritative references as cited.

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